ECEN 248 - Lab Report

Lab #2

A Simple Logic Circuit

Section: 524

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Objectives:
The purpose of this lab is to design and implement a profit calculator for a small farm. We continue to introduce concepts about logical minimization using K-Maps and digital circuits. As we learn about logical minimization, we also learn how to implement DIP switches (or use jumper wires to imitate them) and LEDs in the lab. Throughout the lab, we learn how to design and implement digital circuits by creating truth tables from given constraints, minimizing Boolean expressions with K-Maps, and testing the circuits using these basic logic gates: AND, OR, XOR, and NOT; as well as this lab equipment: breadboard and power supply.
In this farm profit calculator, the four input bits represent different animal types: I (Chicken), S (Sheep), C (Cow), and H (Hog). These inputs are combined to produce a 3-bit binary output (P0, P1, P2), where P2 is the most significant bit. This setup demonstrates the practical application of Boolean algebra and K-map minimization, while also providing hands-on experience with breadboard circuit construction and LED outputs.

Design:
First, we set up the power supply for the experiment. While the power supply is turned off, we connected the power supply leads to the breadboard using only channel one (yellow): red cable connected to the power supply's positive terminal and the pin of this cable to the positive power rail on the breadboard, and the black cable connected to the power supply's negative terminal and the pin of this cable to the negative power rail.

Next, we set up all the gates on the board. We referenced the demo and paper to determine which chips to place on the board. When inserting the chip, we placed it across the center partition of the breadboard and ensured it was positioned correctly to avoid shorting the circuits. Also, always make sure the notch on the edge is on the top side when inserting your chip. Next, for each chip, we set up the VCC and GRND. Using the jumper wires, we connected the positive power rail from the DC power supply to the VCC pin and the negative power rail from the power supply to the GND pin of the chips. All of these placements can be found on the chart that is provided during the lab.

Next, we set up our DIP switches, but instead, we use jumper wires to imitate their use. For each input: S, I, H, and C, we placed the jumper wire on one side connected to the negative rail (for now), and the other on the circuit area of the breadboard (for each input, it has a corresponding row on the board where the other side of the jumper wire is connected).

Then, since we are going to be reading our outputs P_0, P_1, and P_2 from the LED, we have to set these up. We connected one of the pins to the negative power rail (acting as a GRND) and then the other to the row on the row of where it's going to read the output from.

Finally, we will test the circuit. To set up the circuit (for others as well), we put the XOR gate on the top left, AND on the top right, INV on the bottom left, and OR on the bottom right. First, we will implement the output P0 and display it through the LED. Using this equation: P_0 = I + H + CS, set up the circuit accordingly. Then connect the output of this equation to the input for the P_O LED. Test all the gates by doing combinations of HIGH and LOW for S, I, H, and C. Repeat this process for P1 and P2. Make sure to check the power supply is off before changing the inputs for the next combination.

Results:
This is the experimental results from testing the profit calculator. One truth table is made demonstrating the output behavior for each bit: P0, P1, and P2 with 16 input combinations.

S	I	H	C	P0	P1	P2
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	1	1	0
0	0	1	1	1	1	1
0	1	0	0	1	0	0
0	1	0	1	1	1	0
0	1	1	0	1	1	0
0	1	1	1	1	1	0
1	0	0	0	0	1	0
1	0	0	1	1	1	1
1	0	1	0	1	0	1
1	0	1	1	1	0	1
1	1	0	0	1	1	0
1	1	0	1	1	1	0
1	1	1	0	1	0	1
1	1	1	1	1	1	1

Table 1. Truth table for P0, P1, and P2 outputs.

P0, P1, and P2 Equations are:

$$\mathbf{P0 = I + H + CS}$$
$$\mathbf{P1 = H \oplus S + CI}$$
$$\mathbf{P2 = I'C + HS}$$

P0 outputs 1 when any of these conditions are met:

- Chickens are present (I = 1)
- Hogs are present (H = 1)
- Both cows AND sheep are present together (C = 1 AND S = 1)

P0 outputs 0 only when no animals are present, or when only sheep are present, or when only cows are present. This creates a system that activates for chickens alone, hogs alone, or the specific combination of cows with sheep.

P1 outputs 1 under two conditions:

- Exactly one of hogs or sheep is present (mutually exclusive via XOR)
- Both cows AND chickens are present together (CI = 1)

The XOR operation means P1 activates when hogs and sheep appear separately but not together. Additionally, P1 activates whenever both cows and chickens are present, regardless of other animals.

P2 is the most significant bit, outputting 1 only in two specific situations:

- Cows are present without chickens (I'C = 1)
- Both hogs AND sheep are present together (HS = 1)

P2 responds to either a "cows-only" condition or the simultaneous presence of both hogs and sheep. It outputs 0 in the majority of cases (9 out of 16), making it highly restrictive.

Conclusion:

We successfully designed and implemented a farm profit calculator using digital logic circuits with Boolean expressions P0 = I + H + CS, P1 = H ⊕ S + CI, and P2 = I'C + HS. By experimenting with the breadboard, DC power supply, logic gates, and LEDs, we confirmed that each output's behavior matched its expected truth table when given all 16 possible input combinations of the four animal types (chickens, sheep, cows, and hogs). The circuit calculated profit values ranging from 0 to 7 as intended. At the end of this lab, we have gained experience on how to successfully design, implement, and test digital logic circuits using Boolean algebra and K-map minimization techniques with the lab equipment used throughout the experiment

Post-Lab Deliverables:

1. Make a table with 16 rows showing the digital inputs and outputs of your circuit observed during the lab. For input combinations not allowed by the aforementioned guidelines, provide the actual output values observed rather than 'X'.

S	I	H	C	P0	P1	P2
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	1	1	0
0	0	1	1	1	1	1
0	1	0	0	1	0	0
0	1	0	1	1	1	0
0	1	1	0	1	1	0
0	1	1	1	1	1	0
1	0	0	0	0	1	0
1	0	0	1	1	1	1
1	0	1	0	1	0	1
1	0	1	1	1	0	1
1	1	0	0	1	1	0
1	1	0	1	1	1	0
1	1	1	0	1	0	1
1	1	1	1	1	1	1

Table 2: This table demonstrates P0, P1, and P2 with 16 different input combinations.

2. What values does your circuit output for the "don't care" inputs and why?

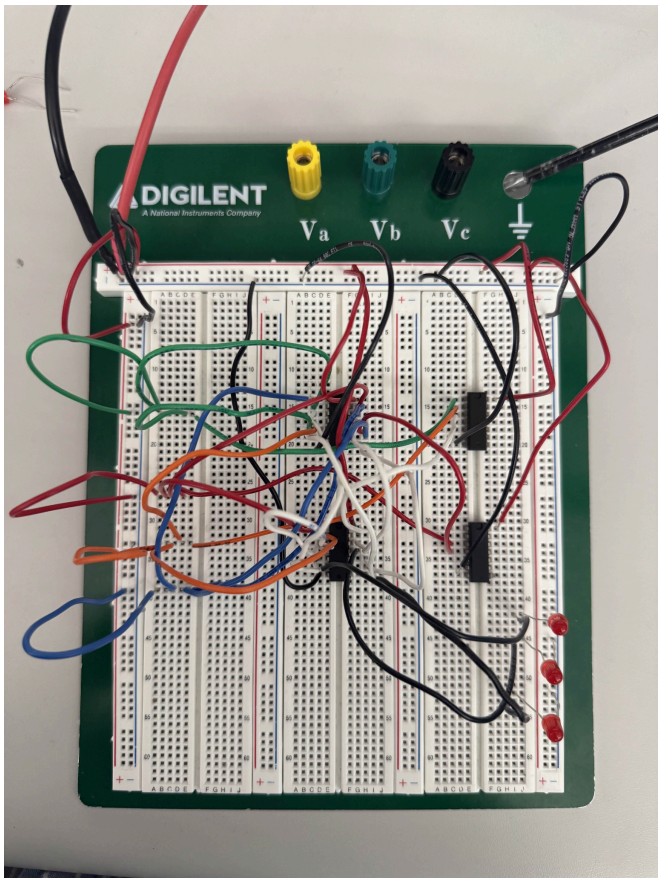
Based on the lab manual, the "don't care" input are combinations that violate the farming guidelines. Here are the rules:

- Rule 1: cannot raise all 4 animals at the same time
- Rule 2: cannot raise more than 2 types of animals simultaneously

So the values that output for the "dont care" inputs include examples like (1, 1, 1, 1), (1, 1, 1, 0), (1, 1, 0, 1), (1, 0, 1, 1), and (0, 1, 1, 1). When tested, these combinations still produced specific binary outputs

because the circuit's Boolean expressions evaluate mathematically without recognizing the farming guidelines. During the K-Map minimization, these were treated as 'X' values to create simpler expressions, but the physical circuit must output actual values for all input combinations (even though this shouldn't happen because of the guidelines).

3. Take a picture of your complete circuit with a seven-segment display. Try your best to introduce your product to John and teach him how to use it.



To use the profit calculator, you must start by setting up the inputs: Chicken (I), Cows (C), Sheep (S) and Hog (H). To set the inputs as present (or 1), you put the input to the positive rail, and negative if you want the input to be not present (0). The profit calculator will output to the LED lights demonstrating the outputs for P0, P1, and P2. P2 will be the most significant put and P0 is the least significant bit. When you want to switch if a certain animal is present or not, just switch the input from the positive to negative rail or vice versa.